

Learning Environments in AR: Comparing Tablet and Head-mounted Augmented Reality Devices at Room and Table Scale

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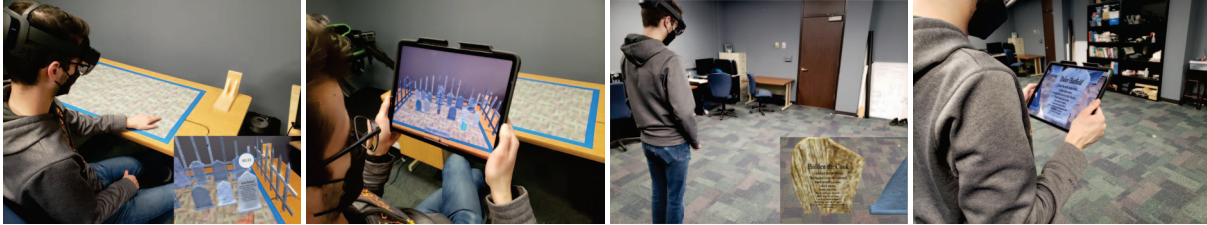


Figure 1: Images from the four conditions: Hololens2 Table Scale, iPad Table Scale, Hololens2 Room Scale, and iPad Room Scale.

ABSTRACT

This paper presents work to examine how presentation scale and device form factor may affect learning in augmented reality (AR) environments. We conducted a 2 (form factor) $\times$ 2 (scale) experiment in which 131 participants explored an AR learning environment using either tablet AR or a Hololens2 crossed with either full room scale or at table scale. Dependent variables measured participants declarative knowledge about information acquired in the environment as well as their understanding of spatial layout. Initial analysis suggests comparable outcomes across all manipulations with respect to acquiring declarative and spatial knowledge.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed/ augmented reality; Human-centered computing—Empirical studies in HCI; Applied computing—Interactive learning environments

1 INTRODUCTION

In this paper we examine spatial and declarative knowledge learning outcomes related to exploring an augmented reality (AR) environment. Theoretically, our work focuses on the affordance of a medium's modality, as defined in the MAIN Model [7]. We build off previous work examining this model with regard to virtual reality (VR) locomotion [4]. Here, we are primarily concerned with two independent variables related to modality: the form factor of an AR device and the impact of the scale of environment. These variables may play into how media and spatial information is acquired while exploring an AR environment. For room-scale, we hypothesized that the ability to walk through a room-scale space would lead to improved spatial understanding of layout. However, large physical spaces for natural locomotion are not always feasible, so a table-scale version of the environment stimuli was created that could be viewed while seated. Currently, AR devices are available in head-mounted and often expensive form factors. This stands in contrast to AR research which use less expensive, handheld devices such as smartphone or tablet AR [1]. Both devices provide a compelling sense of AR and understanding how different devices compare with regard to acquisition of information within an environment resulted in a 2 (Form Factor) by 2 (Scale) experiment. Our dependent variables related

to learning outcomes were assessed by a multiple choice narrative test about information read within the AR environment and a spatial reconstruction test developed to gauge spatial understanding.

2 BACKGROUND AND RELATED WORK

Augmented Reality projects digital elements over a user's view of a real-world environment. Augmented Reality has been considered as an educational tool since the early 1990s [2] In this role AR has been positively received by participants, and has been shown to increase learners engagement [1].

A unique advantage of AR is how it meshes with the real world affording natural interaction. Traditional interfaces require users to learn how to use physical or virtual controls, such as mouse and keyboard or controllers, to rotate and zoom in on virtual objects. Conversely AR allows users to change their view of a virtual object by simply walking around it [3]. The use of natural locomotion over artificial interfaces has already been shown to be beneficial when using VR devices. A 2014 study found that allowing users to physically walk around a virtual environment, improved their results on spatial memory tests, when compared to traditional interfaces such as keyboard and mouse [5]. Within VR environments, users are fully immersed and surrounded by the digital environment. They are often isolated from the physical obstacles or objects in the real environment, which may limit distractions. Interestingly, AR affords users the viewing of both real and digital space with each other. Users have the benefit of a larger spatial presence and may better understand the real-virtual relationships but may also be distracted by these interactions. Understanding the affordances that AR and VR bring to learning environments is the focus of this work.

3 METHOD

We conducted a 2 (Form Factor) by 2 (Scale) between-subjects experiment. Scale was controlled by presenting the stimulus at full room scale or presenting it in miniature so that it would fit on a table top. Two AR form factors investigated either head-mounted AR or tablet AR, resulting in four total conditions, illustrated in Figure 1. Post experiment, participants were tested on declarative learning outcomes using a multiple choice narrative test, and on spatial learning using a spatial reconstruction test.

3.1 Stimulus Material

The visual stimulus material consisted of a virtual cemetery containing 13 tombstones. Each stone was inscribed with a story about the life of one of the inhabitants of the fictional town Spoon River.

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All stories were developed in house and were inspired by the Spoon River Anthology [6]. The combined reading of all tombstones informed a larger story about the inhabitants. The room scale version the cemetery was scaled to fill a 21x33ft lab space. Participants were able to explore this environment by freely walking between tombstones. The table scale environment fit on a 29.5x60in table. Participants viewed the cemetery from above, seated in a chair, and explored by tapping on relevant tombstones.

3.2 Hardware and Software

For our head mounted form factor we used the Microsoft HoloLens2. AR elements are superimposed over a user's field of view using a pair of 2k 3:2 lenses built into the headset. The HoloLens is capable of tracking both the physical environment, and the wearer's hands. In the handheld AR form factor, we used the Apple iPad Pro 12.9. The tablet has a back mounted camera it can use to view the environment. In addition it is equipped with a LiDAR Scanner that enables it to detect physical objects and surfaces. The 3D environment stimulus and software were programmed using Unity with executables targeted for the HoloLens2 and the iPad.

3.3 Experiment procedure

A total of 131 participants were recruited (55 male, 76 female). Participants were university students aged 18-33, with a mean age of 19. Each participant was randomly assigned to one of the four conditions. Before being admitted to the study participants were provided with an informed consent form approved by our university's IRB. Participants were asked to fill out a demographic questionnaire. Self reported information included: sex, GPA, past experience with AR/VR and the number of hours using various digital devices. Before each experiment participants were instructed on how to properly don or hold the AR device and how to use it to interact with the digital elements of the environment. Participants were then walked through a brief training scene to give them practice with the device. After training participants were able to view the cemetery. Participants were directed to visit all tombstones. The sequence of the tombstones was randomized for each participant. A glowing waypoint marked the next tombstone in this sequence. The text on each tombstone was obscured until the waypoint was activated. After activation participants were given 35 seconds to view the text. Once the allotted time was up, the text was hidden and participants were directed to the next tombstone. This process continued until every tombstone had been viewed.

3.3.1 Room Scale and Table Scale Conditions

For the room scale condition participants could freely walk about the lab space to reach waypoints. Participants activated tombstones by standing on the waypoint directly in front of the tombstone.

Participants in the table-scale condition were seated at a 29.5x60 inch table. The cemetery was projected onto the table so that participants viewed it from it's entrance. Participants were asked to stay seated throughout the experiment. To enhance readability upon activation, tombstones would be enlarged and brought forward slightly. In the HoloLens2 condition participants could adjust their view by moving their head in relation to the projection. Tombstones were activated by placing a hand though the stone or it's waypoint marker. In the tablet condition, participants were free to maneuver the iPad to change their view of each tombstone. Tombstones were activated by tapping on the stone or it's waypoint on the iPad screen.

3.4 Evaluation

After exploring the stimulus, participants were given two tests to evaluate their spatial and declarative knowledge learning outcomes. The declarative knowledge test consisted of 20 questions about the narrative presented on all the tombstones. For the spatial test, participants were asked to reconstruct the layout of the cemetery.

The spatial reconstruction task presented the participant with a 2D empty map of the cemetery and asked them to drag images of each tombstone to the correct location on the map. Additional tombstones, not present in the experiment, were added as foils.

	Narrative		Spatial	
	Mean	SD	Mean	SD
HoloLens2-Room	14.1875	3.47746	6.9375	3.49135
HoloLens2-Table	14.5312	4.25012	7.1563	3.49294
iPad-Room	13.0606	4.15286	6.6970	2.88839
iPad-Table	14.0588	3.34792	6.6471	2.75111

Table 1: Results of narrative and spatial tests. A perfect score on the Narrative test is 20 and 13 on the Spatial test.

4 RESULTS

Our results are summarized in Table 1. There was no significant difference in the mean narrative nor spatial assessment scores for each condition. Factorial MANOVA revealed there was neither a main effect for Form Factor, Wilks' $\Lambda = .989$, $F(2, 126) = .713$, $p < .492$, partial $\eta^2 = .011$; nor for Scale, Wilks' $\Lambda = .99$, $F(2, 126) = .635$, $p < .532$, partial $\eta^2 = .01$; nor was there a significant interaction effect between Form Factor and Scale: Wilks' $\Lambda = .996$, $F(2, 126) = .266$, $p < .767$, partial $\eta^2 = .004$.

5 DISCUSSION AND CONCLUSION

A hypothesis going into this experiment was that room-scale conditions may improve learning outcomes related to spatial updating regardless of form factor since participants were able to walk around the full space of the cemetery in AR. Declarative knowledge was hypothesized to be fairly similar across conditions. However, form factor and scale were *both* not significantly different across the two dependent variables discussed in this article. While not statistically different, there may be practical considerations. With a standard grading scale applied to the narrative assessment, the HoloLens2-Table condition results in a 73% (a C) whereas the iPad-Room condition is a 65% (a D). These differences may be related to nuances of modality within AR learning environments and creating spaces that afford users the ability to focus on the content more directly, such as sitting at a table interacting with the stimuli.

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